

* Home Assignment - 2 *

A] Apply appropriate chain rule & solve

2] If $z = \cos(3x - 5y)$ where $x = 4t^4$, $y = \frac{1}{t}$
 find value of $\frac{dz}{dt}$ at $t = 0$

$$z = \cos(3x - 5y)$$

$$\frac{dz}{dt} = -\sin(3x - 5y) \cdot \frac{d}{dt}(3x - 5y)$$

~~cos~~ sin

$$= -\sin(3x - 5y) \cdot \left[3 \frac{dx}{dt} - 5 \frac{dy}{dt} \right] \quad \text{--- (1)}$$

$$\therefore x = 4t^4 = 4 \text{ at } t = 1$$

$$\frac{dx}{dt} = 16t^3 \quad \& \quad y = \frac{1}{t} \quad \therefore \frac{dy}{dt} = -\frac{1}{t^2}$$

$$= -1 \text{ at } t = 1$$

$\therefore$ Substituting, in (1).

$$\frac{dz}{dt} = -\sin(3x - 5y) \left[48t^3 + \frac{5}{t^2} \right]$$

$$\therefore \Rightarrow (3x - 5y)$$

$$\Rightarrow (12 + 6)$$

$$\Rightarrow 18$$

$$\therefore \frac{dz}{dt} = -\sin(18) [48 + 5]$$

$$= -\sin(18) \cdot 53$$

$$= -53 \cdot \sin(18)$$

→

$$z = \cos(3x - 5y)$$

$$x = 4t^4$$

$$y = \frac{1}{t}$$

$$\therefore z = \cos\left(3(4t^4) - 5\left(\frac{1}{t}\right)\right)$$

$$= \cos\left(12t^4 - \frac{5}{t}\right)$$

$$\therefore \frac{dz}{dt} = -\sin\left(12t^4 - \frac{5}{t}\right) \cdot \frac{d}{dt}\left(12t^4 - \frac{5}{t}\right)$$

$$= -\sin\left(12t^4 - \frac{5}{t}\right) \cdot \left(48t^3 + \frac{5}{t^2}\right)$$

$$\therefore \text{at } t=0,$$

$$\therefore \left(12t^4 - \frac{5}{t}\right) \rightarrow -\infty$$

$$12t^4 \rightarrow 0 \quad \& \quad -\frac{5}{t} \rightarrow -\infty$$

$$\therefore \left(12t^4 - \frac{5}{t}\right) \rightarrow -\infty$$

$$\therefore \frac{dz}{dt} = \sin(-\infty)$$

$$\therefore \text{at } t=0, \text{ the val. of } \frac{dz}{dt} \text{ does not exist.}$$

B]

4] If $u = \sin^{-1} \left(\frac{x^3 y^3 z^3}{x^3 + y^3 + z^3} \right)$ eval. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$

→

$$\text{let } v = \frac{x^3 y^3 z^3}{x^3 + y^3 + z^3}$$

$$\therefore u = \sin^{-1}(v)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-v^2}} \frac{\partial v}{\partial x}$$

similarly,

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{1-v^2}} \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial z} = \frac{1}{\sqrt{1-v^2}} \frac{\partial v}{\partial z}$$

$$\begin{aligned} \therefore \frac{\partial v}{\partial x} &= \frac{(x^3 + y^3 + z^3)(3x^2 y^3 z^3) - (x^3 y^3 z^3)(3x^2)}{(x^3 + y^3 + z^3)^2} \\ &= \frac{3x^2 y^3 z^3 (x^3 + y^3 + z^3) - 3x^5 y^3 z^3}{(x^3 + y^3 + z^3)^2} \\ &= \frac{3x^2 y^3 z^3 (y^3 + z^3)}{(x^3 + y^3 + z^3)^2} \end{aligned}$$

similarly,

$$\frac{\partial v}{\partial y} = \frac{3y^2 x^3 z^3 (x^3 + z^3)}{(x^3 + y^3 + z^3)^2} \quad \& \quad \frac{\partial v}{\partial z} = \frac{3z^2 x^3 y^3 (x^3 + y^3)}{(x^3 + y^3 + z^3)^2}$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{1}{\sqrt{1-v^2}} \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} \right)$$

$$= \frac{3}{(x^3 + y^3 + z^3)^2} \left(x^3 y^3 z^3 \left(\frac{y^3 + z^3}{x^3} + \frac{x^3 + z^3}{y^3} + \frac{x^3 + y^3}{z^3} \right) \right)$$

$$\therefore x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = v$$

$$\therefore \frac{6x^3 y^3 z^3}{\sqrt{(x^3 + y^3 + z^3)^2 - (x^3 y^3 z^3)^2}} = \frac{6v}{\sqrt{1-v^2}}$$

c) Find maxima / minima & also maximum & minimum values.

3] $f(x, y) = 2xy - 5x^2 - 2y^2 + 4x + 4y - 4$

$$f_x = \frac{\partial f}{\partial x} = 2y - 10x + 4 = 0 \quad \text{--- (1)}$$

$$f_y = \frac{\partial f}{\partial y} = 2x - 4y + 4 = 0 \quad \text{--- (2)}$$

$$\therefore y = 5x - 5$$

Substituting in eqⁿ (2),

$$\therefore 2x - 4(5x - 5) + 4 = 0$$

$$\Rightarrow 18x = 12$$

$$\therefore x = \frac{2}{3} //$$

$$\therefore y = 5\left(\frac{2}{3}\right) - 5 \therefore y = \frac{4}{3} //$$

$$\therefore \text{Critical pt.} \Rightarrow \left(\frac{2}{3}, \frac{4}{3}\right) //$$

$$\therefore f_{xx} = -10$$

$$\& f_{yy} = -4$$

$$\& f_{xy} = 2 \left(\frac{\partial^2 f}{\partial x \partial y} \right)$$

$$\therefore D = (-10)(-4) - (2)^2 = 36$$

$$\therefore D > 0 \quad \& \quad \frac{\partial^2 f}{\partial x^2} < 0 \quad \therefore \text{critical pt is local max.}$$

$$\begin{aligned} \therefore f\left(\frac{2}{3}, \frac{4}{3}\right) &= 2\left(\frac{2}{3}\right)\left(\frac{4}{3}\right) - 5\left(\frac{2}{3}\right)^2 - 2\left(\frac{4}{3}\right)^2 + 4\left(\frac{2}{3}\right) + 4\left(\frac{4}{3}\right) - 4 \\ &= \frac{16}{9} - \frac{20}{9} - \frac{32}{9} + \frac{24}{9} + \frac{48}{9} - \frac{36}{9} \end{aligned}$$

$$= \frac{0}{9} = 0 // = \text{max. value.}$$